Shape of data

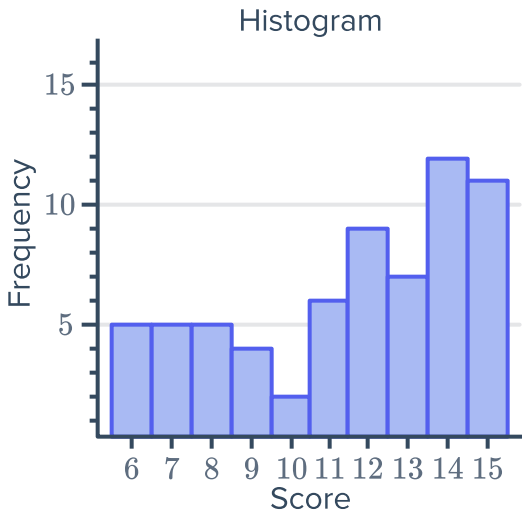
1 Describe the shape of the data in the following graphs:

a

	Leaf
1	6 7 7
2	2 2 2 2 3 3 3
3	3 3 3 6 6 6 7 7 7 7 7
4	4 4 4 4 4 4
5	7 7

Key: $2|3 = 23$

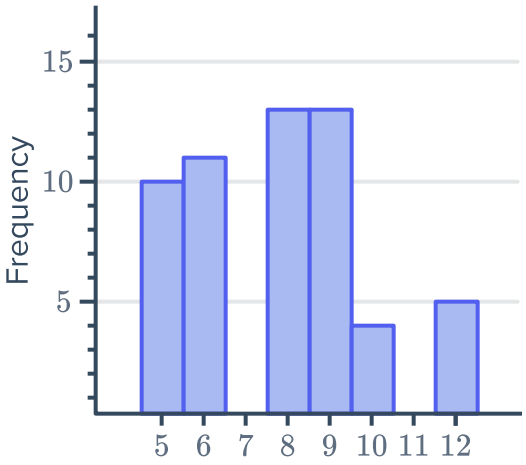
b



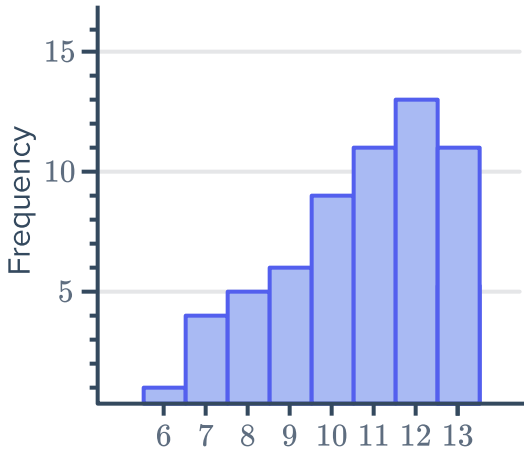
c



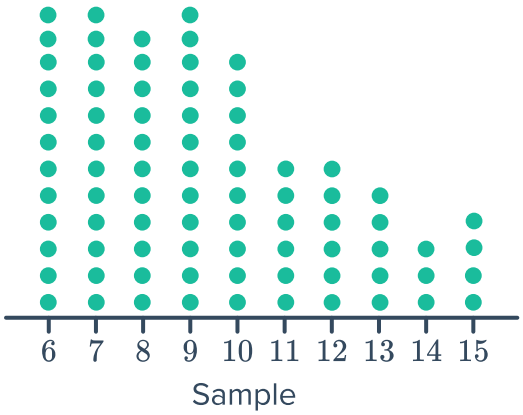
d



e

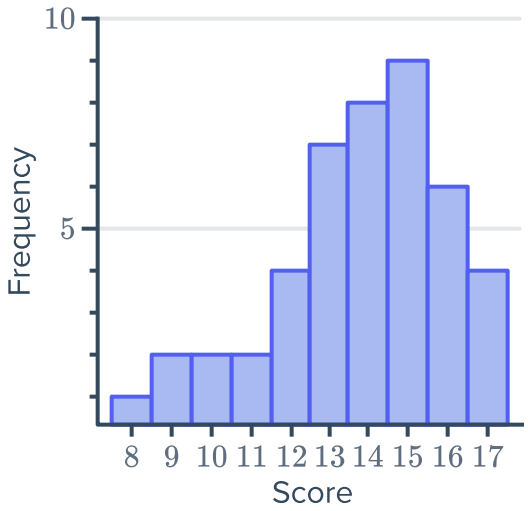


f

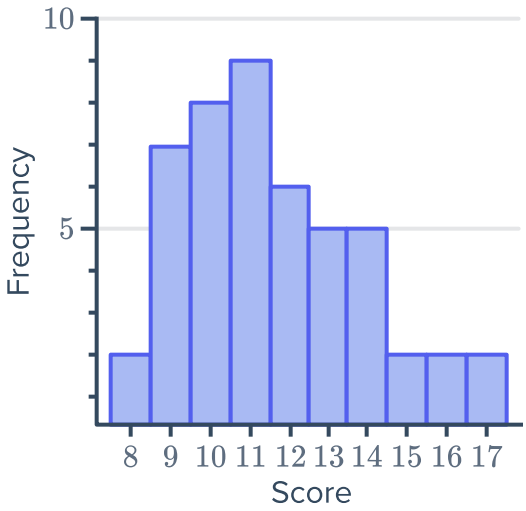


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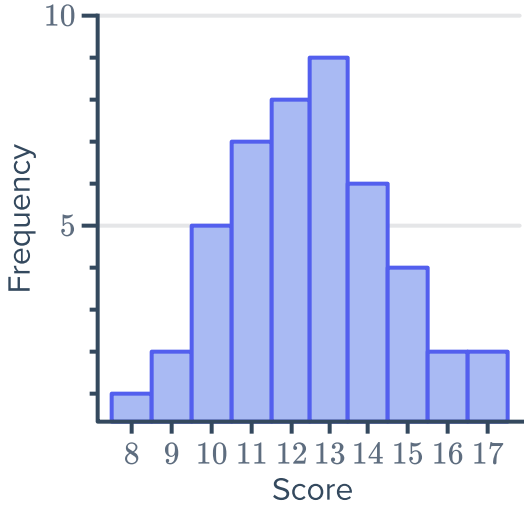
g



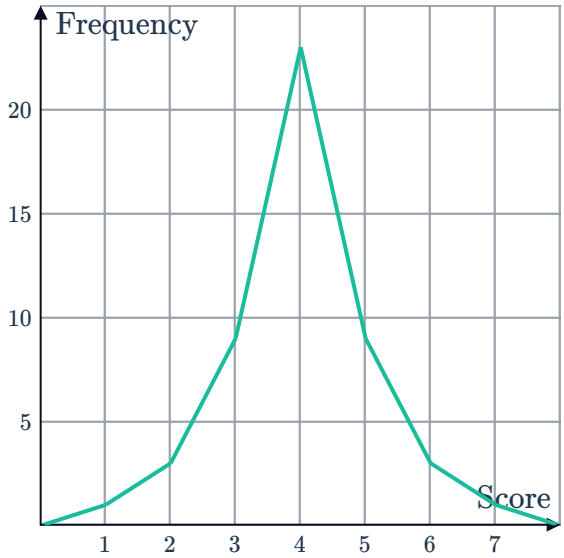
h



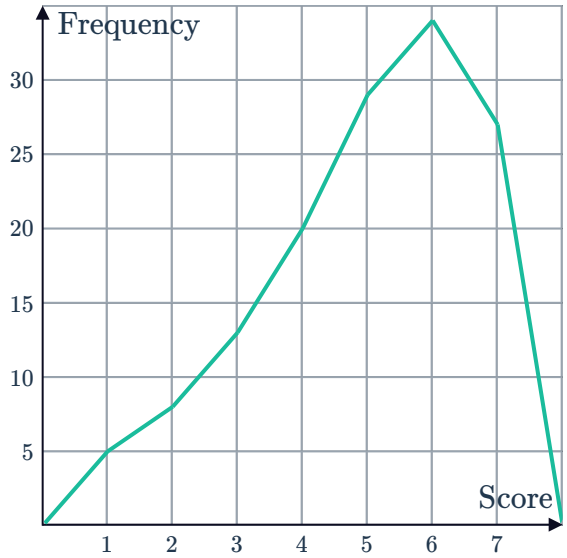
i



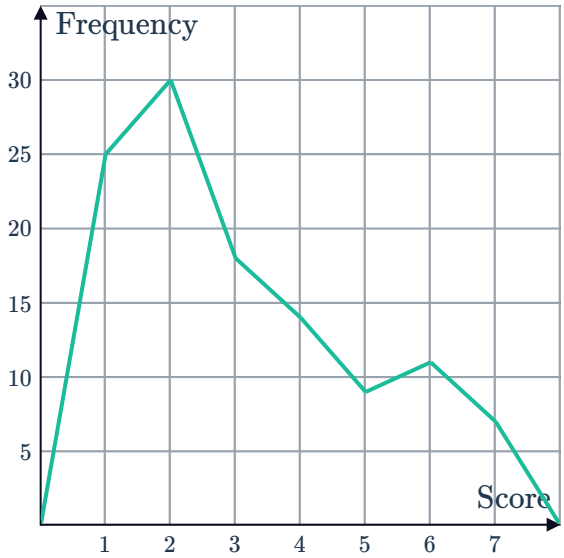
j



k

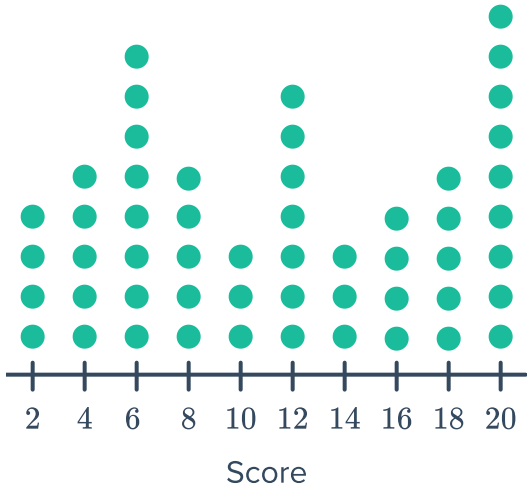


l



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2 How would you describe the modality of the following dot plot? Explain your answer.



3 The table shows the number of crime novels in a bookshop for different price ranges rounded off to the nearest $5:

a Graph this data in a histogram.

b Describe the shape of the distribution of the data.

Price of crime novel	Frequency
$5	5
$10	10
$15	17
$20	8
$25	17
$30	10
$35	5

Clusters and outliers

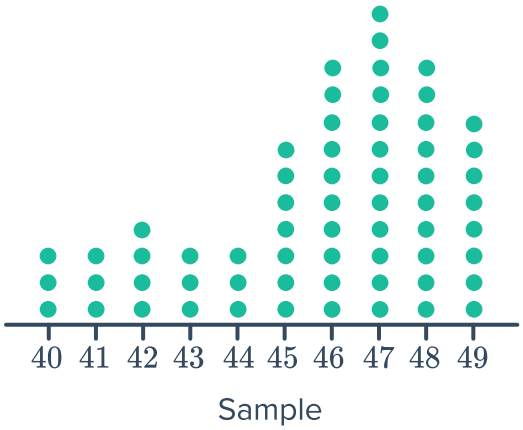
4 Temperatures were recorded over a period of time and presented as a dot plot:

a Are there any outliers?

b Is there any clustering of data? If so, in what interval?

c What is the modal temperature?

d Describe the shape of the distribution.



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5 The number of hours worked per week by a group of people is represented in the following stem and leaf plot:

	Leaf
0	2
1	
2	0 3 6 6
3	1 4 5 6 6 7
4	0 4 6 7 9
5	0

Key: $2|3 = 23$

a State the value of any outliers.

b Is there any clustering of data? If so, in what interval?

c State the mode.

6 Consider the stem plot given:

	Leaf
0	5
1	7 8
2	0 8
3	1 3 3 7 8 9
4	1 3 5 8 8 8
5	
6	
7	
8	
9	2

Key: $2|3 = 23$

a State the value of any outliers.

b Is there any clustering of data? If so, in what interval?

c State the mode.

d Describe the shape of the distribution.

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7 Consider the dot plot below:



a Are there any outliers?

b Is there any clustering of data?

c State the modal score(s).

d Describe the shape of the distribution.

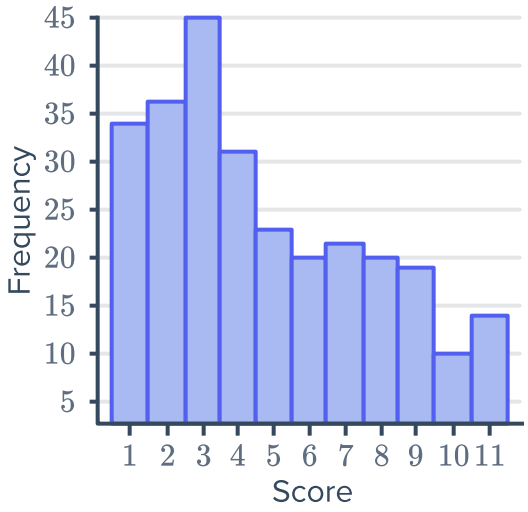
8 Consider the data shown in the histogram:

a Are there any outliers?

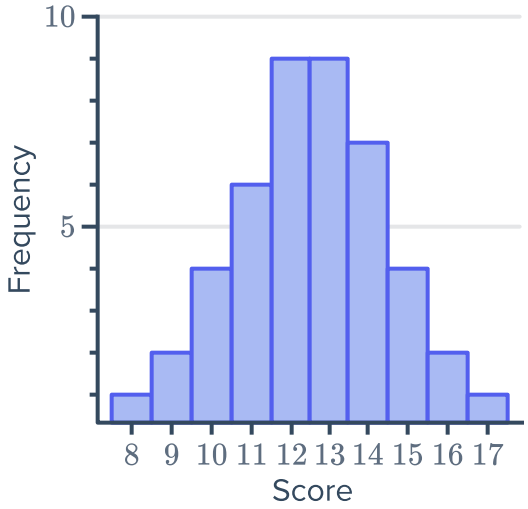
b Is there any clustering of data? If so, in what interval?

c State the mode.

d Describe the shape of the distribution.

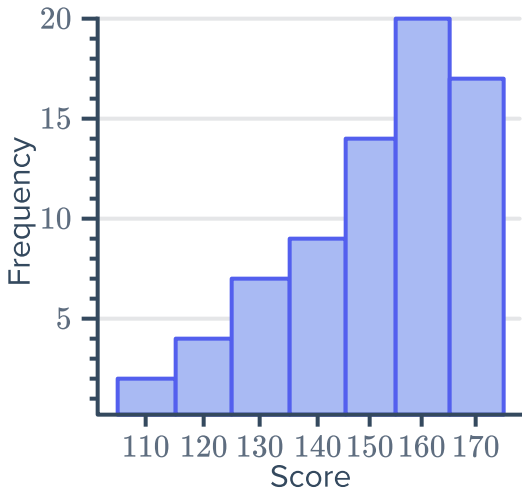


9 Estimate the value of the mean of the following data set correct to one decimal place:



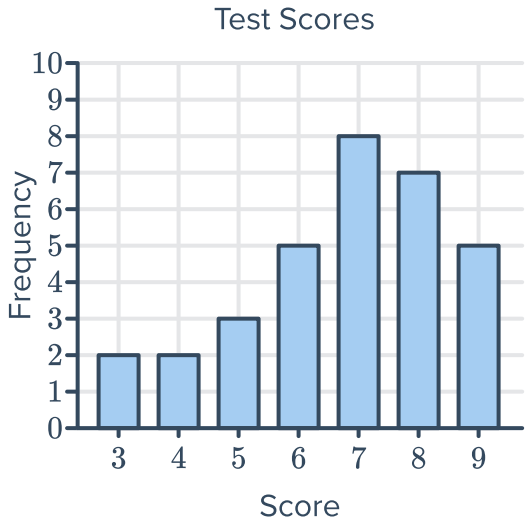
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10 Consider the histogram representing students' heights in centimetres:



- **a** Does the histogram most likely represent grouped data or individual scores?
- **b** Estimate the value of the mean to one decimal place.
- **c** Describe the shape of the distribution.

11 Consider the given column graph:



- **a** Describe the shape of the distribution.
- **b** Find the following:
  - **i** Lower quartile
  - **ii** Upper quartile
- **c** Calculate the interquartile range.
- **d** Are there any outliers? If so, state the value.

12 Consider the dot plot given:



- **a** Describe the shape of the distribution.
- **b** Find the following:
  - **i** Lower quartile
  - **ii** Upper quartile
- **c** Calculate the interquartile range.
- **d** Are there any outliers? If so, state the value.

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13 The stem and leaf plot below shows the age of people to enter through the gates of a concert in the first 5 seconds:

a Find the median age.

b Find the difference between the least age and the median.

c Find the difference between the greatest age and the median.

d Calculate the mean age, correct to two decimal places.

e Is the data positively or negatively skewed?

	Leaf
1	0 1 2 2 3 3 4 4 4 8 8 8
2	1 7
3	4 5 5
4	0
5	4

Key: $1|2 = 12$ years old

14 $\mathrm{VO_2Max}$ is a measure of how efficiently your body uses oxygen during exercise. The more physically fit you are, the greater your $\mathrm{VO_2Max}$. Here are some people's results, listed in ascending order, when their $\mathrm{VO_2Max}$ was measured:

$$21,\ 21,\ 23,\ 25,\ 26,\ 27,\ 28,\ 29,\ 29,\ 29,\ 30,\ 30,\ 32,\ 38,\ 38,\ 42,\ 43,\ 44,\ 48,\ 50,\ 76$$

a Determine the median $\mathrm{VO_2Max}$.

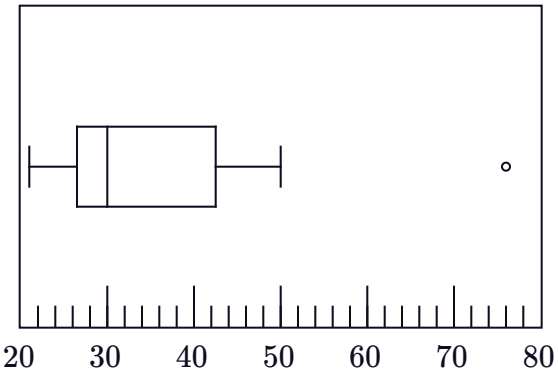
b Determine the upper quartile value.

c Determine the lower quartile value.

d Consider the box plot for this data set and state whether the results are positively or negatively skewed.

e State the value of the outlier.

f An average untrained healthy person has a $\mathrm{VO_2Max}$ between 30 and 40. What can we say about the exercise habits of the majority of this group of people?

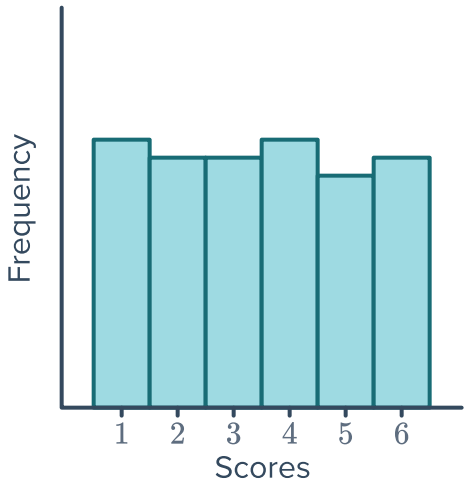


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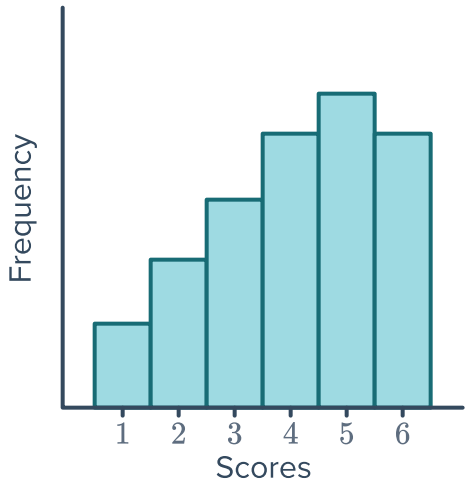
15 A die is rolled for a large number of trials and the number appearing is noted.

a Which histogram would you expect to match the data? Explain your answer.

A



B



b Describe the shape of the distribution of the data chosen above.

16 A pair of dice are rolled and the numbers appearing on the uppermost face are added to create a score.

a How many combinations would result in a score of:

i 4 **ii** 7 **iii** 10

b A pair of dice are rolled for a large number of trials and the numbers appearing are added to create a score. Draw a sketch of a histogram you would expect to match the data.

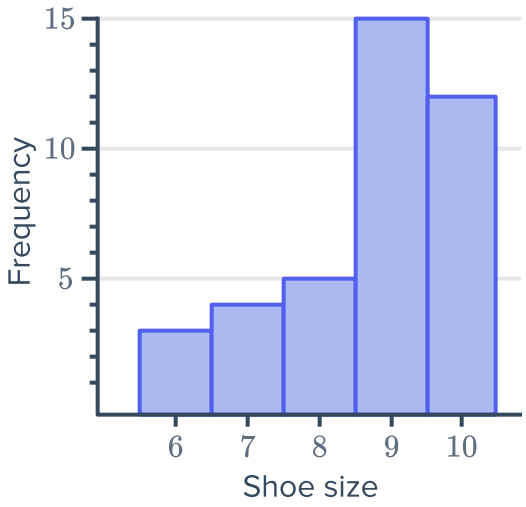
17 The shoe sizes of all the students in a class were measured and the data was presented in a graph:

a Are there any outliers?

b Is there any clustering of data? If so, in what interval?

c What is the modal shoe size?

d Describe the shape of the distribution.

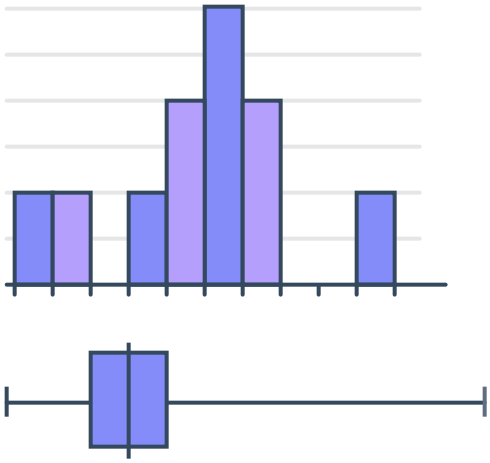


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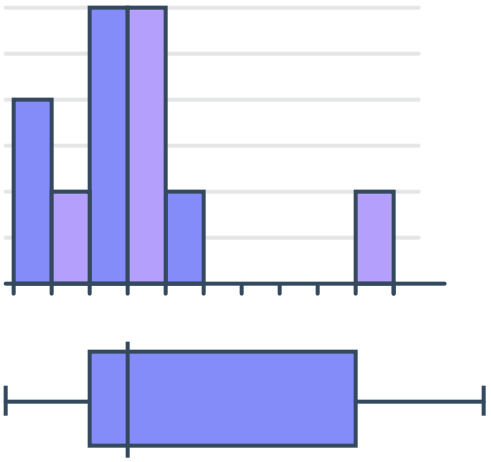
Histograms and box plots

18 Explain why the following pairs of histograms and box plots do not match:

a



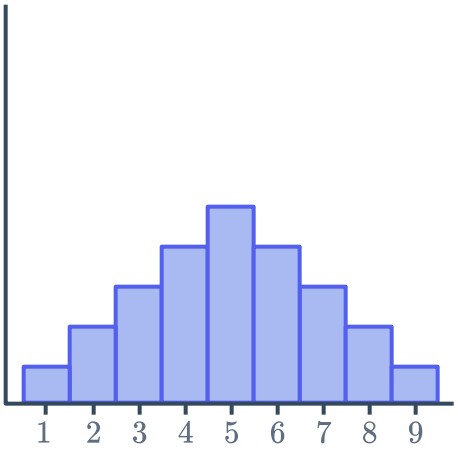
b



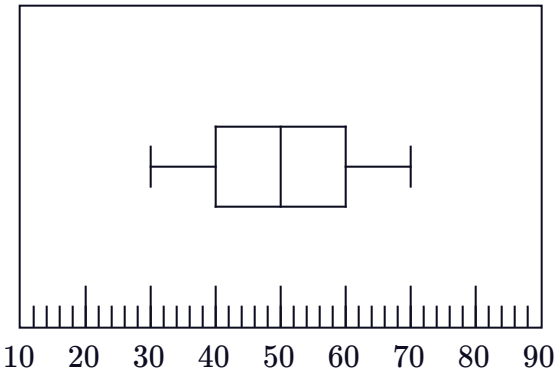
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19 Match the histograms on the left to the corresponding box plots on the right:

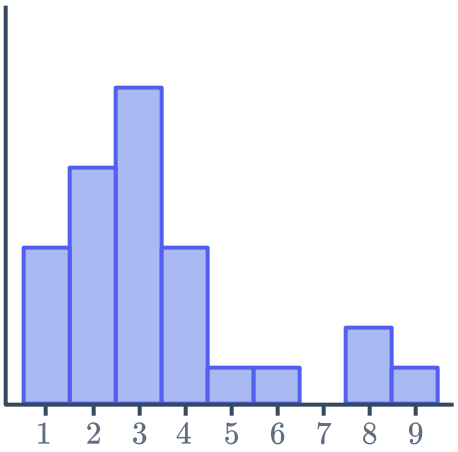
Histogram A



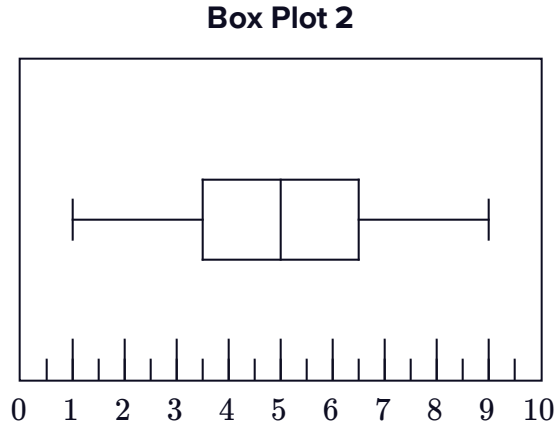
Box Plot 1



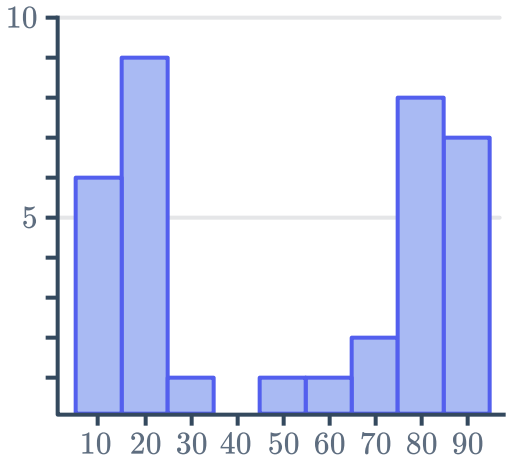
Histogram B



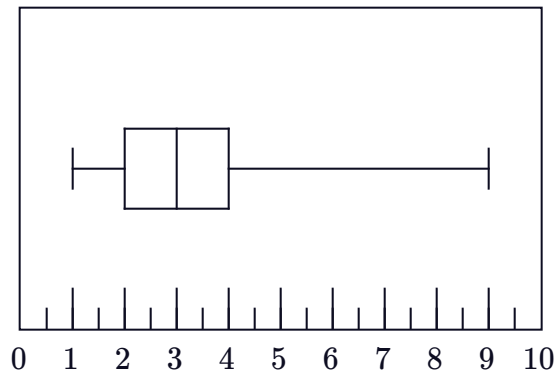
Box Plot 2



Histogram C

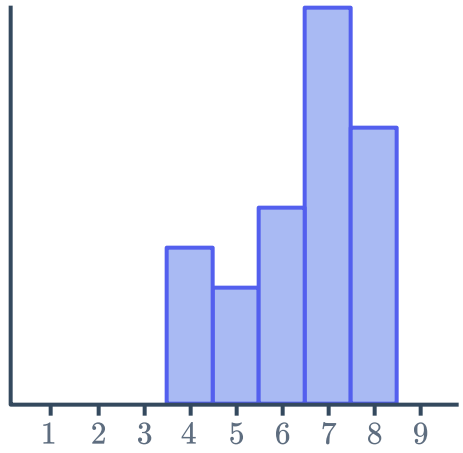


Box Plot 3

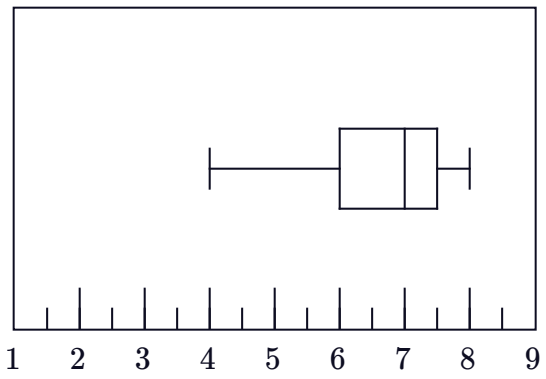


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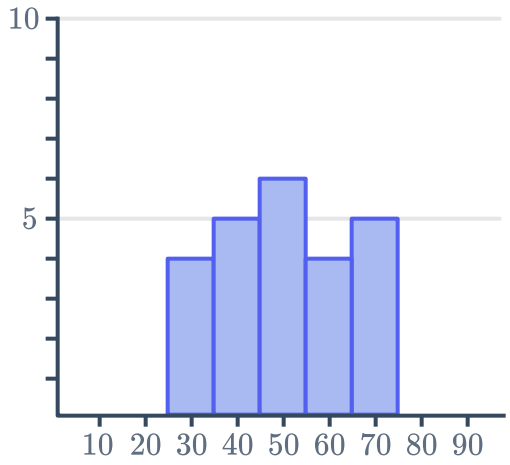
Histogram D



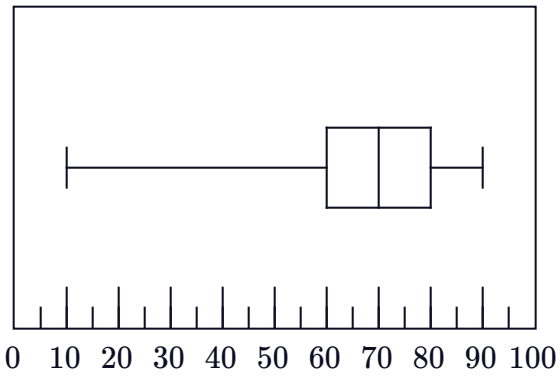
Box Plot 4



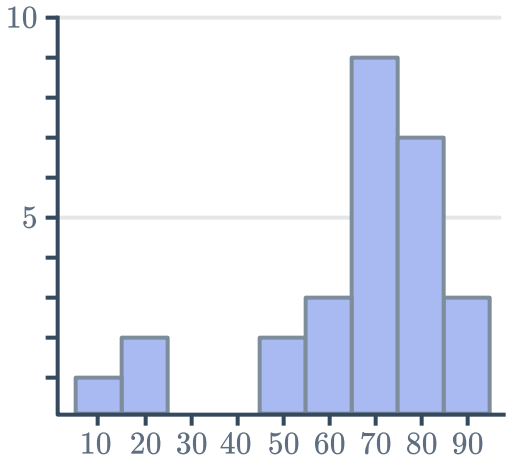
Histogram E



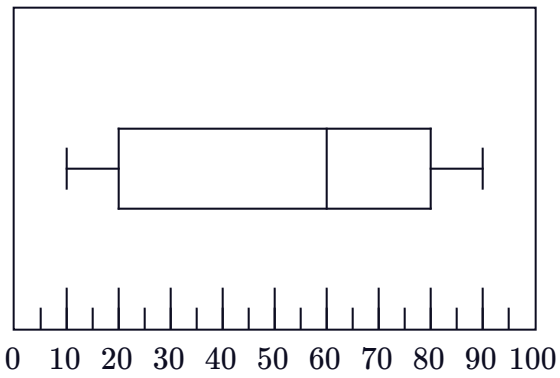
Box Plot 5



Histogram F



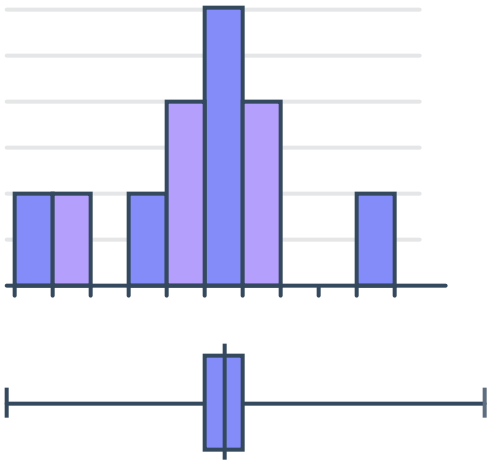
Box Plot 6



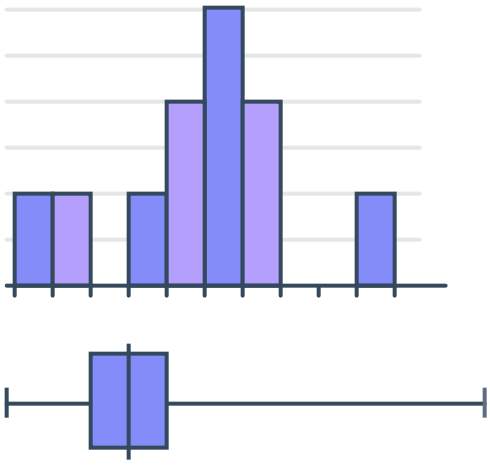
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20 State whether the following pairs of histograms and box plots are correctly matched with respect to their shape:

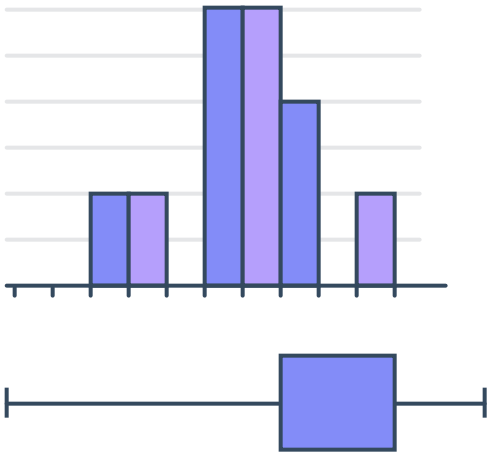
a



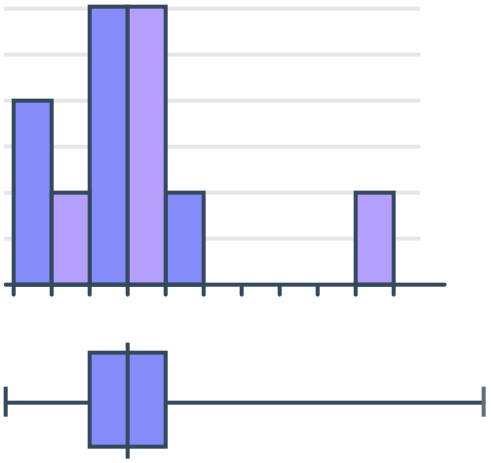
b



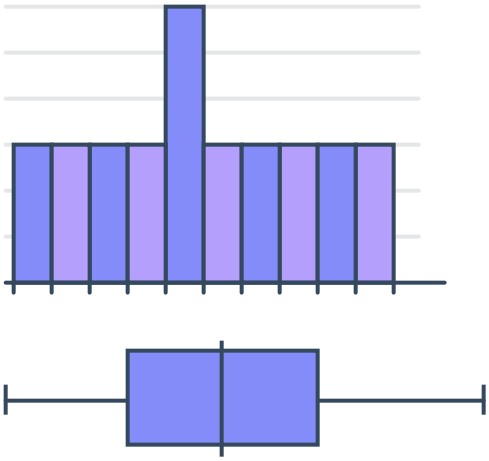
c



d



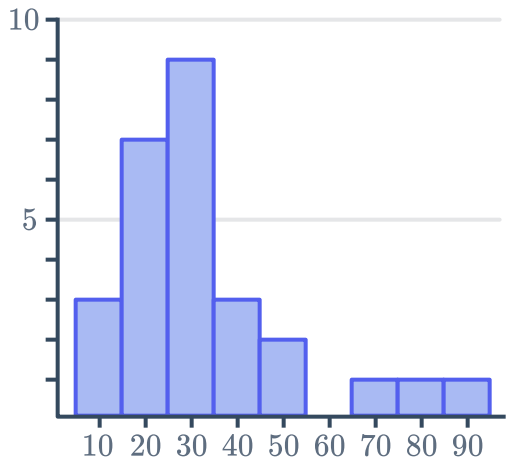
e



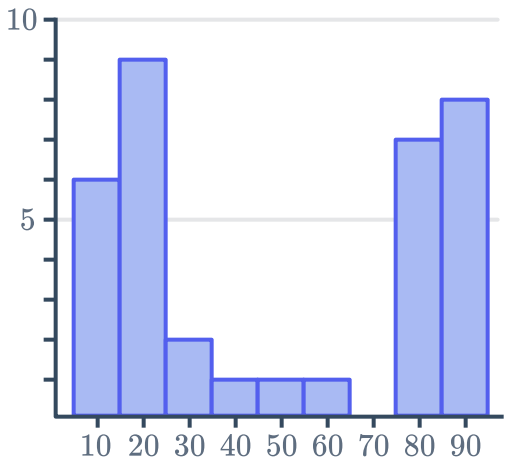
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21 Construct a box plot for the following histograms:

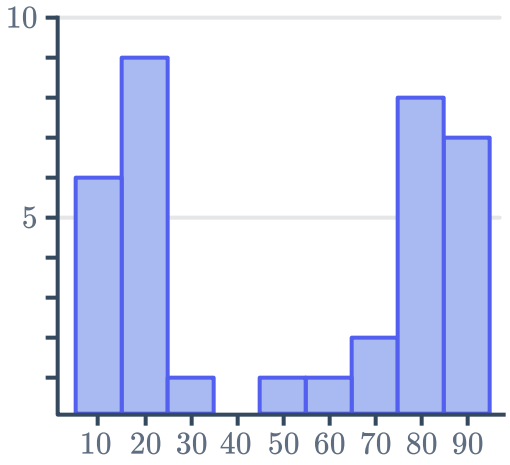
a



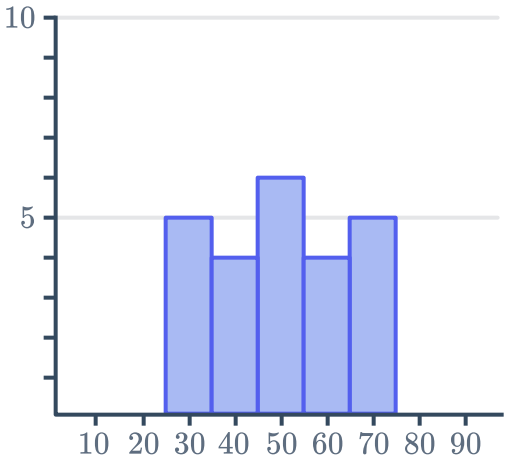
b



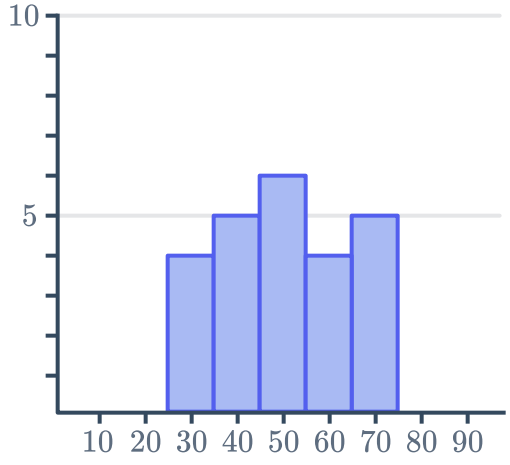
c



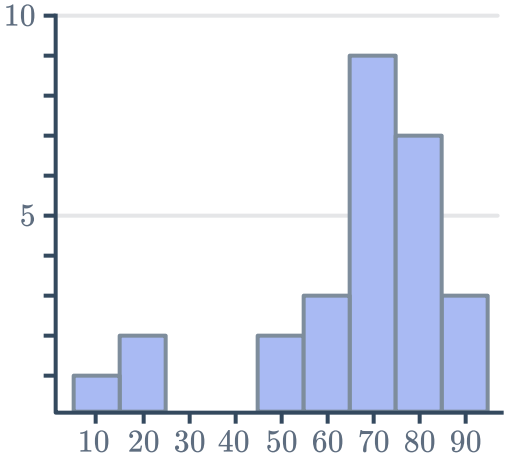
d



e



f



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